

## UNIT 4 - LAB EXPERIMENTS

### EXPERIMENT NO. 1

1. Consider that you are owning a supermarket mall and through membership cards, you have some basic data about your customers like Customer ID, age, gender, annual income and spending score. For the above scenario, the Problem Statement was You want to understand the customers who can easily converge [Target Customers] so that the data can be given to the marketing team and plan the strategy accordingly. For the above scenario prepare a dataset and perform **Clustering Analysis** to segment the customers in the Mall. There are clearly Five segments of Customers based on their Annual Income and Spending Score namely *Usual Customers*, *Priority Customers*, *Senior Citizen* *Target Customers*, and *Young Target Customers*. Sample data

|   | CustomerID | Gender | Age | Annual Income (k\$) | Spending Score (1-100) |
|---|------------|--------|-----|---------------------|------------------------|
| 0 | 1          | Male   | 19  | 15                  | 39                     |
| 1 | 2          | Male   | 21  | 15                  | 81                     |
| 2 | 3          | Female | 20  | 16                  | 6                      |
| 3 | 4          | Female | 23  | 16                  | 77                     |
| 4 | 5          | Female | 31  | 17                  | 40                     |

@relation mall\_customer\_segmentation

@attribute CustomerID numeric

@attribute Gender {Male,Female}

@attribute Age numeric

@attribute AnnualIncome numeric

@attribute SpendingScore numeric

@data

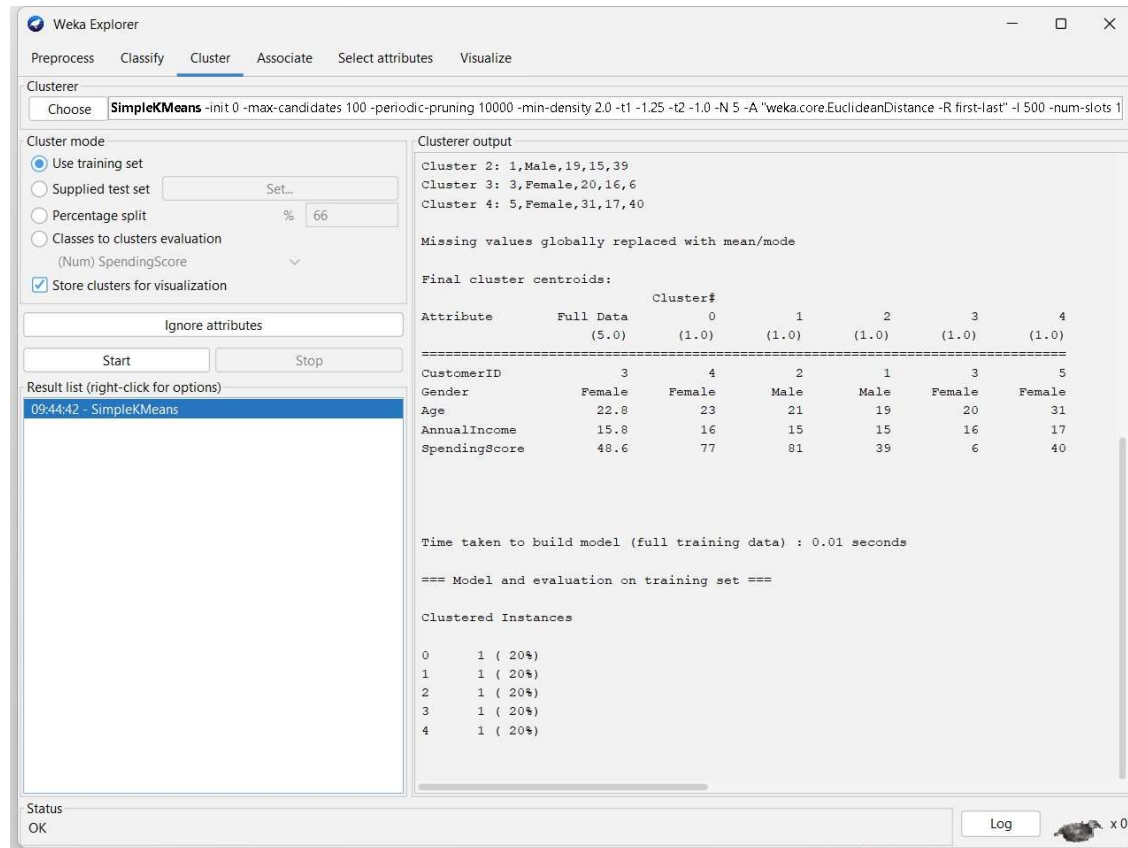
1, Male, 19, 15, 39

2, Male, 21, 15, 81

3, Female, 20, 16, 6

4,Female,23,16,77

5,Female,31,17,40



## EXPERIMENT NO.: 2

2.Create the following dataset using CSV file format. To perform cluster analysis using K-Means in WEKA. To change the cluster size and plot the graph and illustrate the visualization of cluster.

| EmployeeID | Gender | Age | Salary | Credit |
|------------|--------|-----|--------|--------|
| 111        | Male   | 28  | 150000 | 39     |
| 222        | Male   | 25  | 150000 | 27     |
| 333        | Female | 26  | 160000 | 42     |
| 444        | Female | 25  | 160000 | 40     |
| 555        | Female | 30  | 170000 | 64     |
| 666        | Male   | 29  | 200000 | 72     |

@relation employee\_cluster\_analysis

@attribute EmployeeID numeric

@attribute Gender {Male,Female}

@attribute Age numeric

@attribute Salary numeric

@attribute Credit numeric

@data

111,Male,28,150000,39

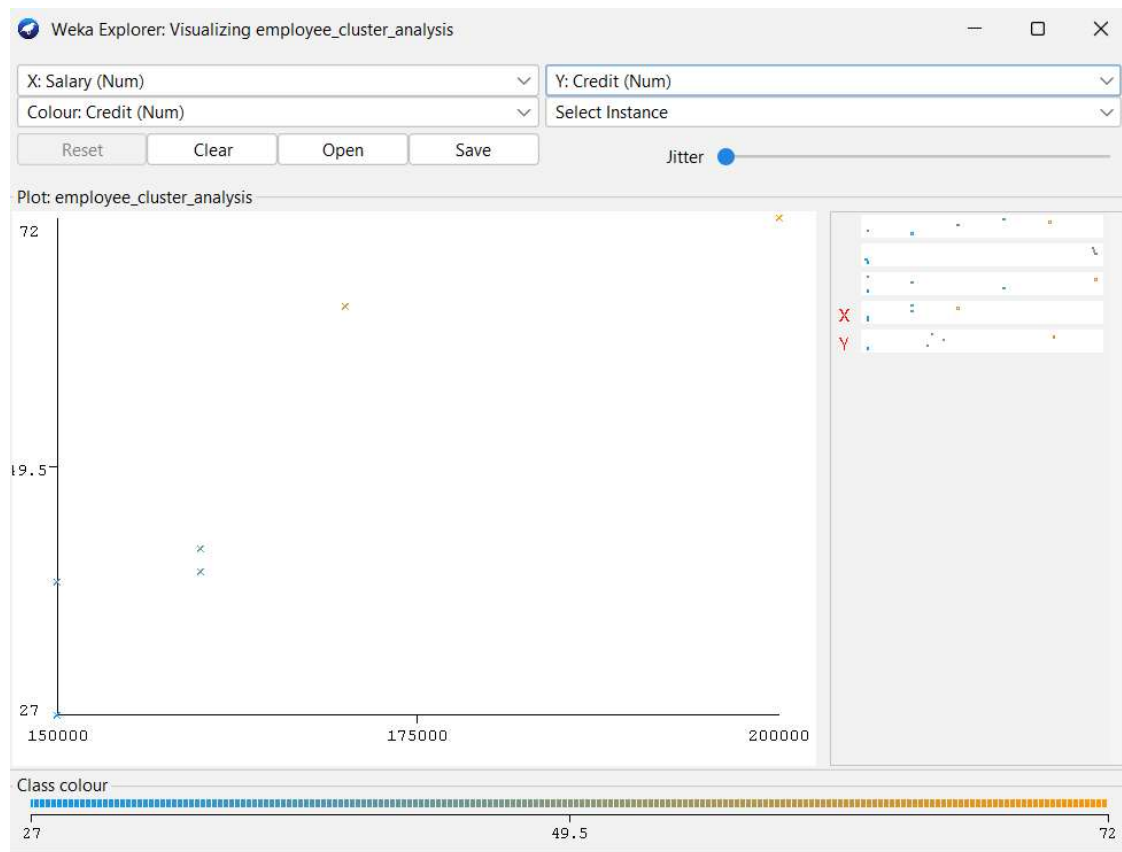
222,Male,25,150000,27

333,Female,26,160000,42

444,Female,25,160000,40

555,Female,30,170000,64

666,Male,29,200000,72



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Credit

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

09:44:42 - SimpleKMeans

09:47:14 - SimpleKMeans

Clusterer output

Cluster 2: 222, Male, 25, 150000, 27

Cluster 3: 555, Female, 30, 170000, 64

Cluster 4: 333, Female, 26, 160000, 42

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute  | Full Data<br>(6.0) | Cluster#<br>0<br>(2.0) | 1<br>(1.0) | 2<br>(1.0) | 3<br>(1.0) | 4<br>(1.0) |
|------------|--------------------|------------------------|------------|------------|------------|------------|
| EmployeeID | 388.5              | 388.5                  | 111        | 222        | 666        | 555        |
| Gender     | Male               | Female                 | Male       | Male       | Male       | Female     |
| Age        | 27.1667            | 25.5                   | 28         | 25         | 29         | 30         |
| Salary     | 165000             | 160000                 | 150000     | 150000     | 200000     | 170000     |
| Credit     | 47.3333            | 41                     | 39         | 27         | 72         | 64         |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 33%) |
| 1 | 1 ( 17%) |
| 2 | 1 ( 17%) |
| 3 | 1 ( 17%) |
| 4 | 1 ( 17%) |

Status OK

Log x0

### EXPERIMENT NO.: 3

3. Prediction of categorical data using Naïve Bayes classification through WEKA using any datasets. Compare the Naïve Bayes algorithm with SVM using the summary of results given by the classifiers and plot the graph.

@relation student\_performance

@attribute StudyHours {low, medium, high}

@attribute Attendance {poor, average, good}

@attribute Result {pass, fail}

@data

low, poor, fail

low, average, fail

medium, average, pass

medium,good,pass

high,good,pass

high,average,pass

low,good,fail

medium,poor,fail

high,poor,pass

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseNaiveBayes

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) Result

StartStop

Result list (right-click for options)

09:51:04 - bayes.NaiveBayes

Classifier output

Scheme:weka.classifiers.bayes.NaiveBayes

Relation:student\_performance

Instances:9

Attributes:3

StudyHours

Attendance

Result

Test mode:10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

| Attribute  | Class  |        |
|------------|--------|--------|
|            | pass   | fail   |
|            | (0.55) | (0.45) |
| =====      |        |        |
| StudyHours |        |        |
| low        | 1.0    | 4.0    |
| medium     | 3.0    | 2.0    |
| high       | 4.0    | 1.0    |
| [total]    | 8.0    | 7.0    |
|            |        |        |
| Attendance |        |        |
| poor       | 2.0    | 3.0    |
| average    | 3.0    | 2.0    |
| good       | 3.0    | 2.0    |
| [total]    | 8.0    | 7.0    |

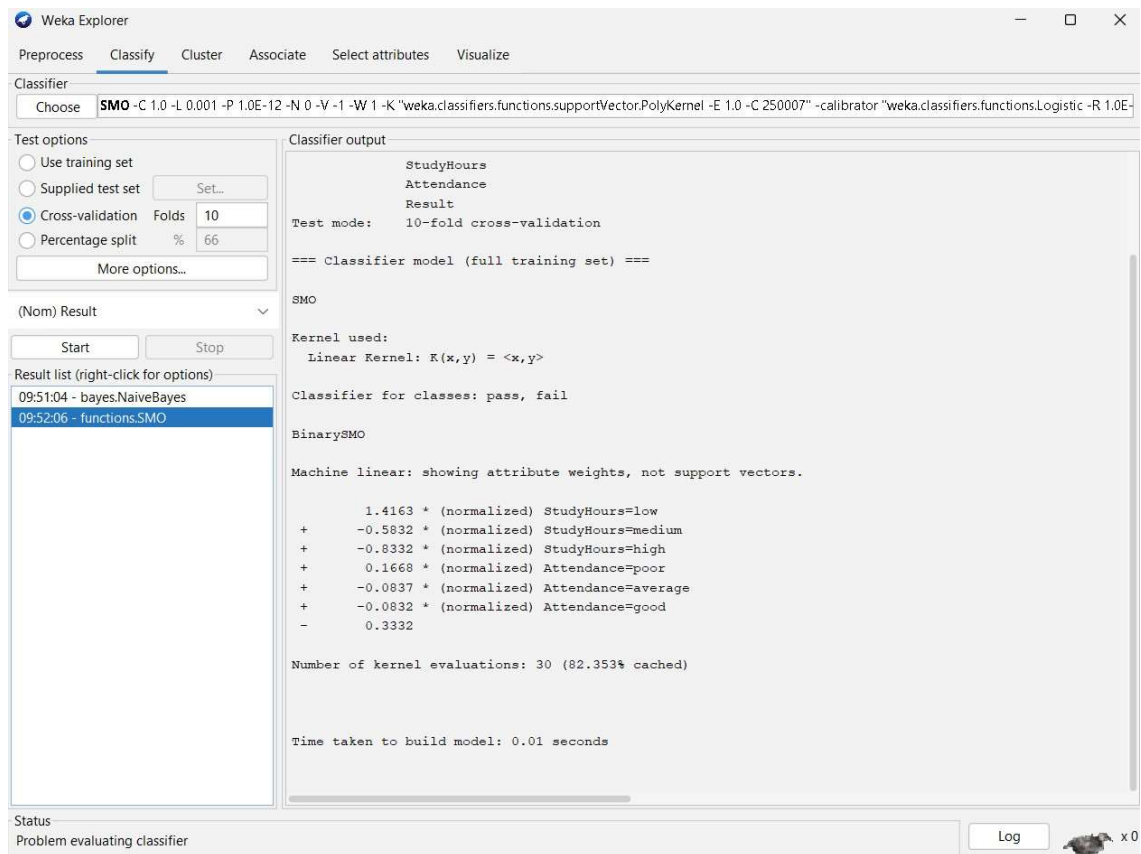
Time taken to build model: 0 seconds

Status

Problem evaluating classifier

Log

x0



## EXPERIMENT NO.: 4

4.The following list of persons with vegetarian or not details given in the table. How will you find out how many of them are vegetarian and how many of them are non-vegetarian? Which type of the person total count is greater value?

| Person     | Gopu | Babu | Baby | Gopal | Krishna | Ja | De | Malin | Hem | Anu |
|------------|------|------|------|-------|---------|----|----|-------|-----|-----|
| Vegetarian | yes  | yes  | yes  | no    | yes     | no | no | yes   | yes | yes |

@relation vegetarian\_analysis

@attribute Person string

@attribute Vegetarian {yes,no}

@data

Gopu,yes  
Babu,yes  
Baby,yes  
Gopal,no  
Krishna,yes  
Jai,no  
Dev,no  
Malini,yes  
Hema,yes  
Anu,yes

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose ZeroR

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Vegetarian

Start Stop

Result list (right-click for options)

09:51:04 - bayes.NaiveBayes

09:52:06 - functions.SMO

09:54:54 - rules.ZeroR

Classifier output

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

|                                  |        |    |   |
|----------------------------------|--------|----|---|
| Correctly Classified Instances   | 7      | 70 | % |
| Incorrectly Classified Instances | 3      | 30 | % |
| Kappa statistic                  | 0      |    |   |
| Mean absolute error              | 0.4727 |    |   |
| Root mean squared error          | 0.5012 |    |   |
| Relative absolute error          | 100    | %  |   |
| Root relative squared error      | 100    | %  |   |
| Total Number of Instances        | 10     |    |   |

=== Detailed Accuracy By Class ===

|               | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC | ROC Area | PRC Area | Class |
|---------------|---------|---------|-----------|--------|-----------|-----|----------|----------|-------|
|               | 1.000   | 1.000   | 0.700     | 1.000  | 0.824     | ?   | 0.000    | 0.700    | yes   |
|               | 0.000   | 0.000   | ?         | 0.000  | ?         | ?   | 0.000    | 0.300    | no    |
| Weighted Avg. | 0.700   | 0.700   | ?         | 0.700  | ?         | ?   | 0.000    | 0.580    |       |

=== Confusion Matrix ===

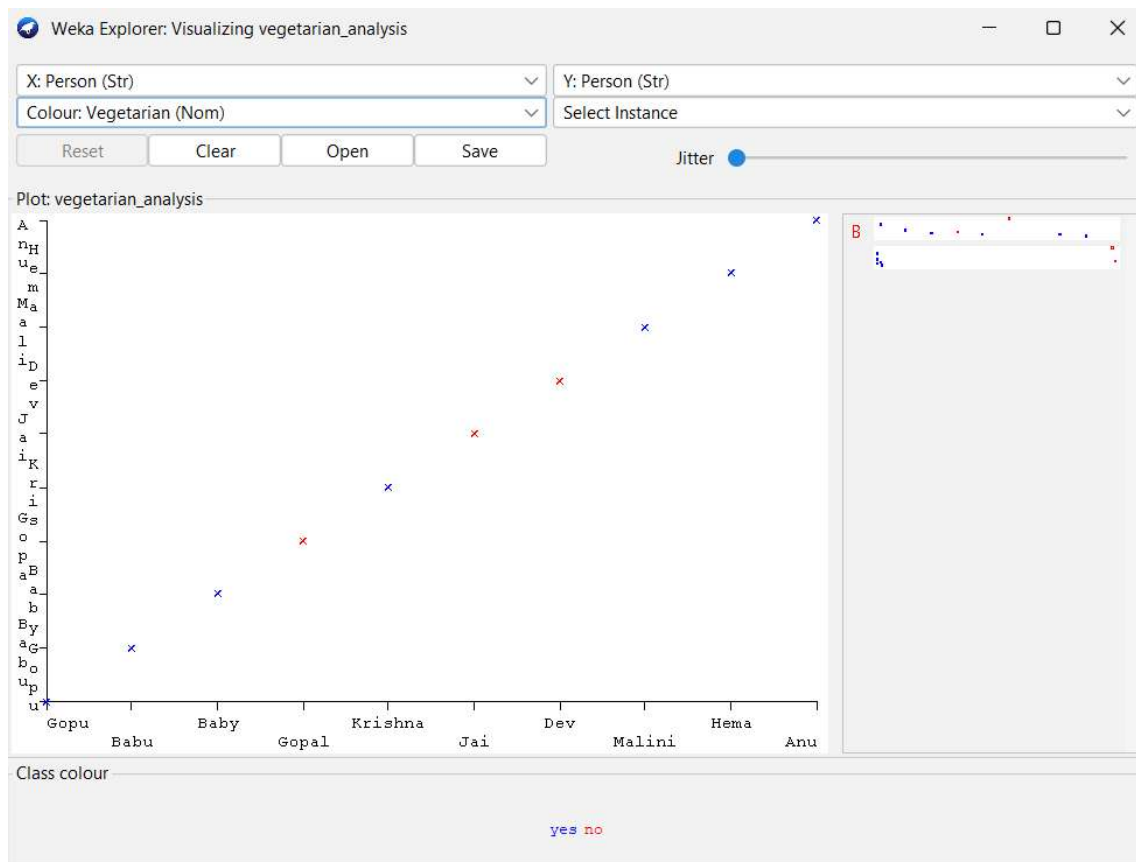
a b <-- classified as

7 0 | a = yes

3 0 | b = no

Status OK

Log x 0



## EXPERIMENT NO.: 5

5.The following table would be plotted as (x,y) points, with the first column being the x values as number of mobile phones sold and the second column being the y values as money. To use the scatter plot for how many mobile phones sold.

|   |    |   |    |    |    |   |     |    |     |     |
|---|----|---|----|----|----|---|-----|----|-----|-----|
| x | 4  | 1 | 5  | 7  | 10 | 2 | 50  | 25 | 90  | 36  |
| y | 12 | 5 | 13 | 19 | 31 | 7 | 153 | 72 | 275 | 110 |

@relation mobile\_sales

@attribute PhonesSold numeric

@attribute Money numeric

@data

4,12

1,5



5,13

7,19

10,31

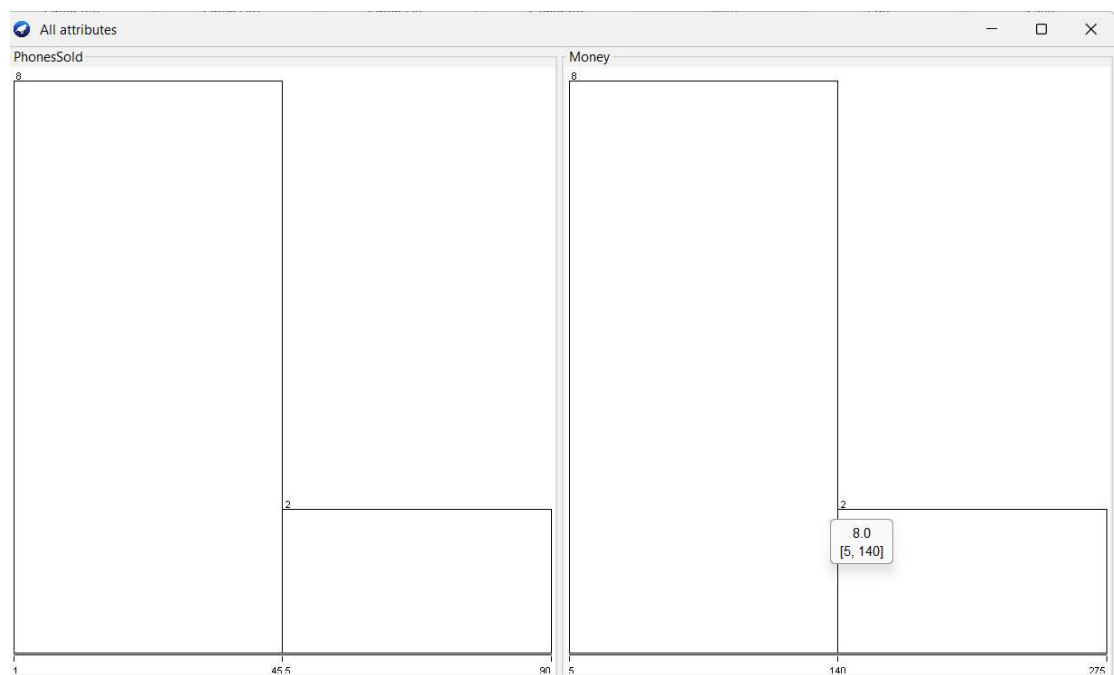
2,7

50,153

25,72

90,275

36,110



## EXPERIMENT NO.: 6

6. Generate rules using FP growth algorithm using the given dataset which has the following transactions with items purchased: Consider the values as support=50% and confidence=75%.

| Transaction ID | Items Purchased           |
|----------------|---------------------------|
| 1              | Bread, Cheese, Egg, Juice |
| 2              | Bread, Cheese, Juice      |
| 3              | Bread, Milk, Yogurt       |
| 4              | Bread, Juice, Milk        |
| 5              | Cheese, Juice, Milk       |

@relation items

@attribute Bread {t,f}

@attribute Cheese {t,f}

@attribute Egg {t,f}

@attribute Juice {t,f}

@attribute Milk {t,f}

@attribute Yogurt {t,f}

@data

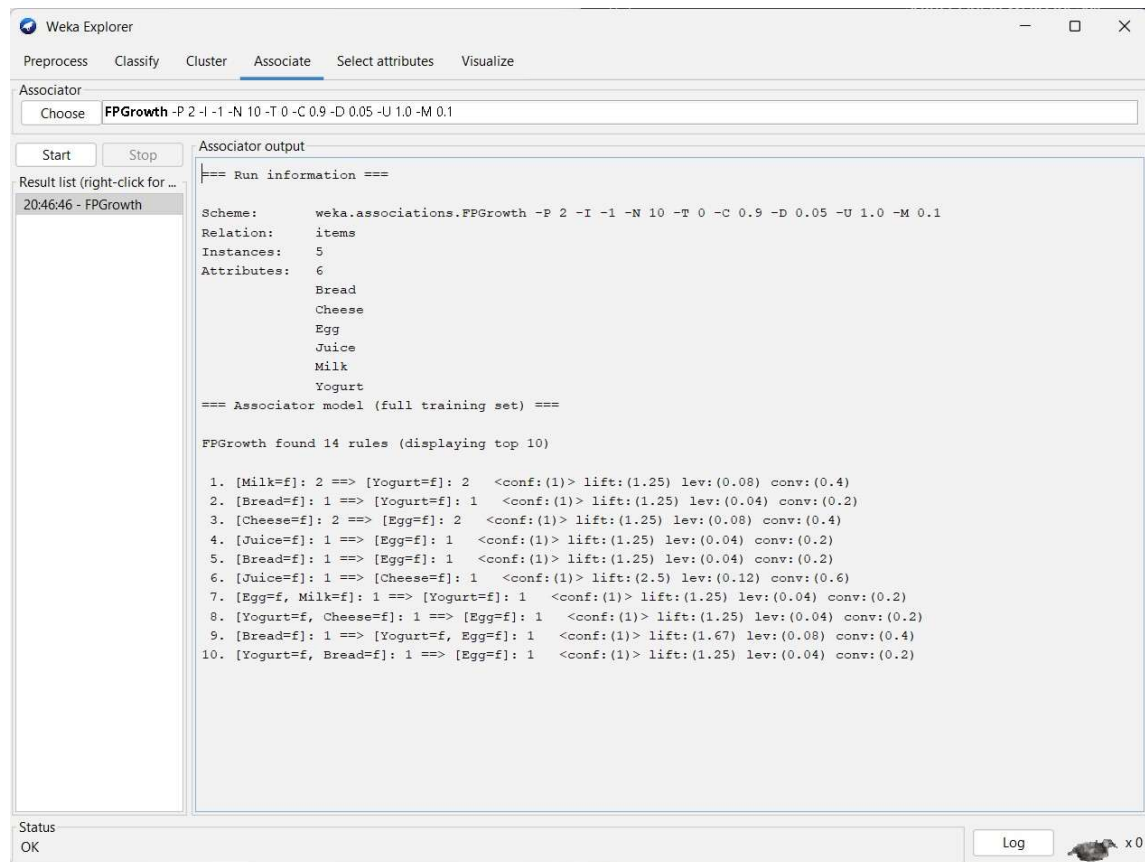
t,t,t,t,f,f

t,t,f,t,f,f

t,f,f,f,t,t

t,f,f,t,t,f

f,t,f,t,t,f



EXPERIMENT NO.:7

@relation diabetes

@attribute Pregnancies numeric

@attribute Glucose numeric

@attribute BloodPressure numeric

@attribute SkinThickness numeric

@attribute Insulin numeric

@attribute BMI numeric

@attribute DiabetesPedigreeFunction numeric

@attribute Age numeric

@attribute Outcome {tested\_positive, tested\_negative}

@data

6,148,72,35,0,33.6,0.627,50,tested\_positive

1,85,66,29,0,26.6,0.351,31,tested\_negative

8,183,64,0,0,23.3,0.672,32,tested\_positive

1,89,66,23,94,28.1,0.167,21,tested\_negative

0,137,40,35,168,43.1,2.288,33,tested\_positive

5,116,74,0,0,25.6,0.201,30,tested\_negative

3,78,50,32,88,31.0,0.248,26,tested\_positive

10,115,0,0,0,35.3,0.134,29,tested\_negative

2,197,70,45,543,30.5,0.158,53,tested\_positive

8,125,96,0,0,0.0,0.232,54,tested\_negative

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier Choose J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Outcome

Start Stop

Result list (right-click for options)

21:41:38 - trees\_J48

Classifier output

Number of Leaves : 2

Size of the tree : 3

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

|                                  |            |    |   |
|----------------------------------|------------|----|---|
| Correctly Classified Instances   | 7          | 70 | % |
| Incorrectly Classified Instances | 3          | 30 | % |
| Kappa statistic                  | 0.4        |    |   |
| Mean absolute error              | 0.38       |    |   |
| Root mean squared error          | 0.5621     |    |   |
| Relative absolute error          | 69.6667 %  |    |   |
| Root relative squared error      | 103.0588 % |    |   |
| Total Number of Instances        | 10         |    |   |

=== Detailed Accuracy By Class ===

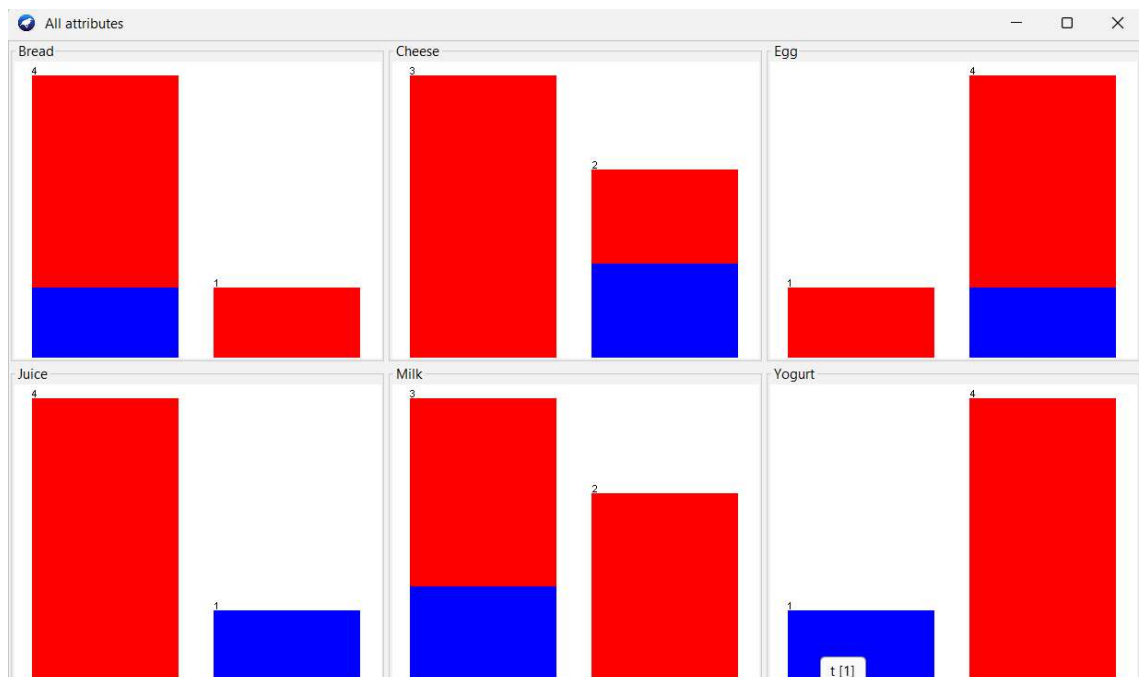
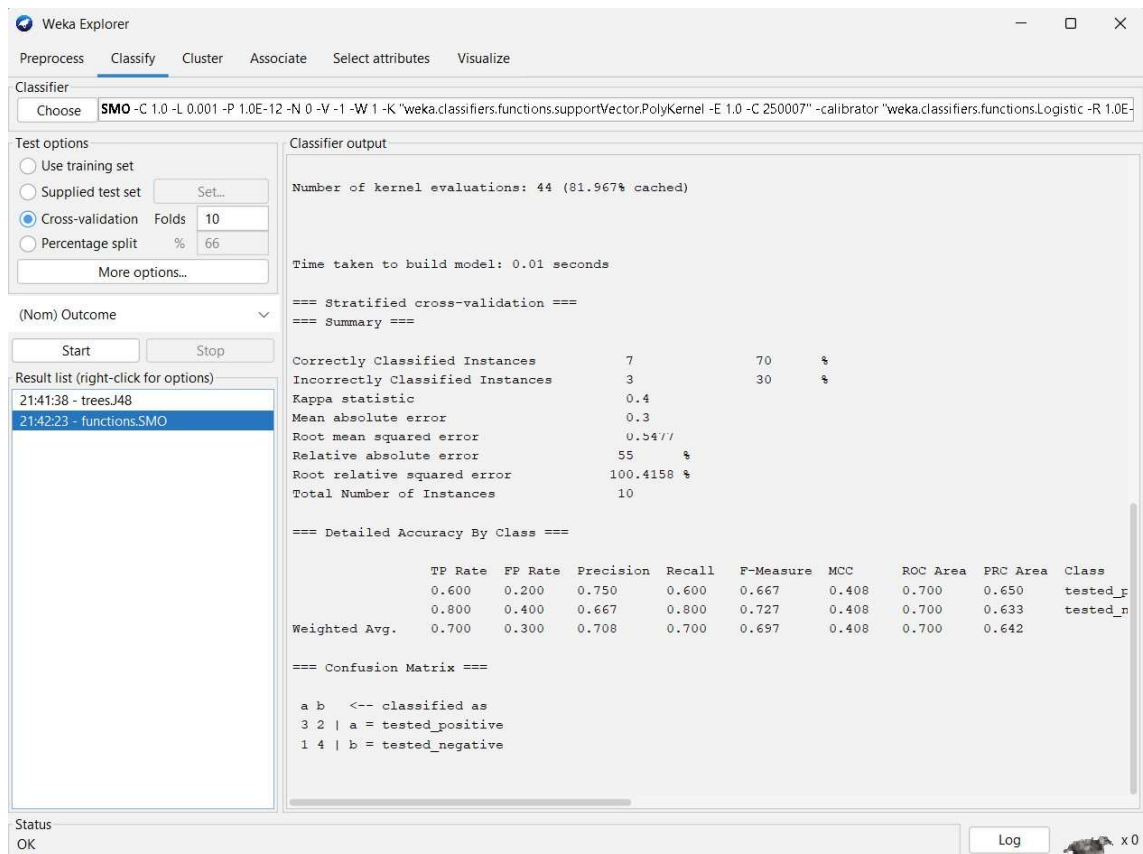
|               | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC   | ROC Area | PRC Area | Class    |
|---------------|---------|---------|-----------|--------|-----------|-------|----------|----------|----------|
|               | 0.600   | 0.200   | 0.750     | 0.600  | 0.667     | 0.408 | 0.540    | 0.650    | tested_p |
|               | 0.800   | 0.400   | 0.667     | 0.800  | 0.727     | 0.408 | 0.540    | 0.633    | tested_n |
| Weighted Avg. | 0.700   | 0.300   | 0.708     | 0.700  | 0.697     | 0.408 | 0.540    | 0.642    |          |

=== Confusion Matrix ===

```
a b <-- classified as
3 2 | a = tested_positive
1 4 | b = tested_negative
```

Status OK

Log



EXPERIMENT NO.:8

@relation marks\_partition

@attribute Marks numeric

@data

55

60

71

63

55

65

50

55

58

59

61

63

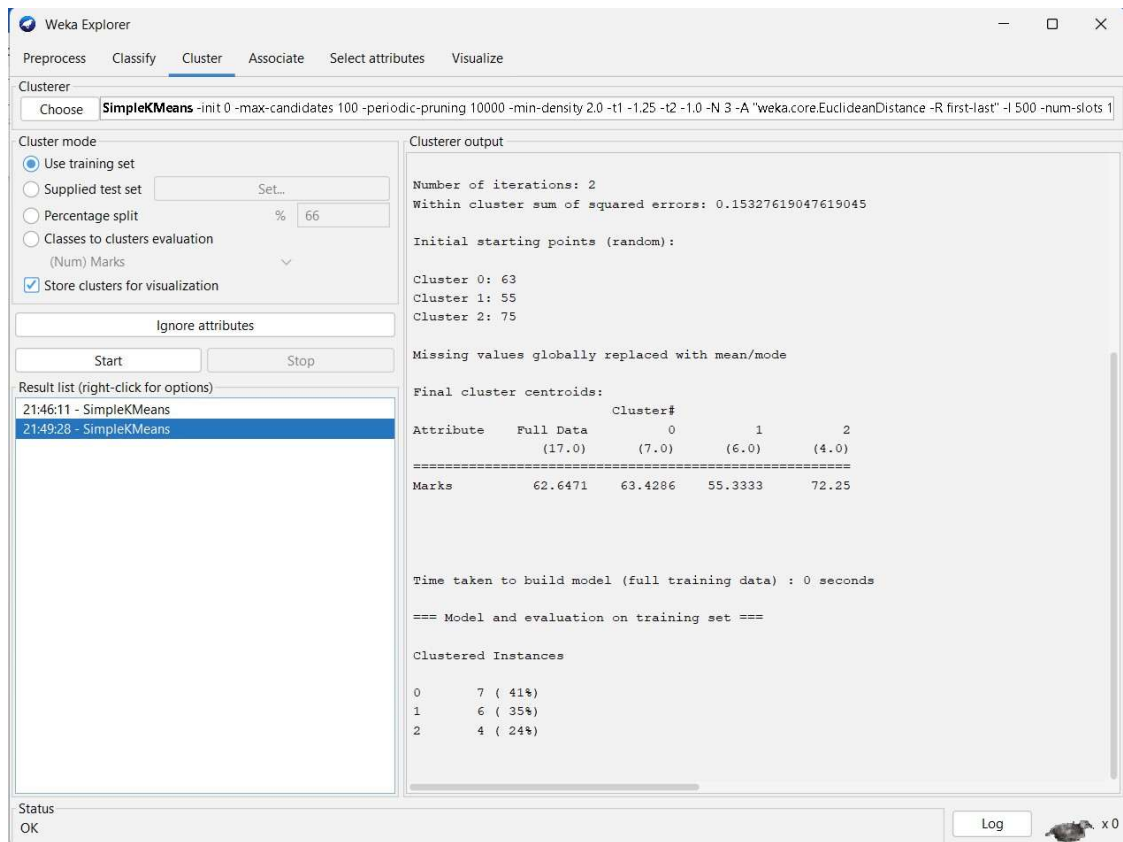
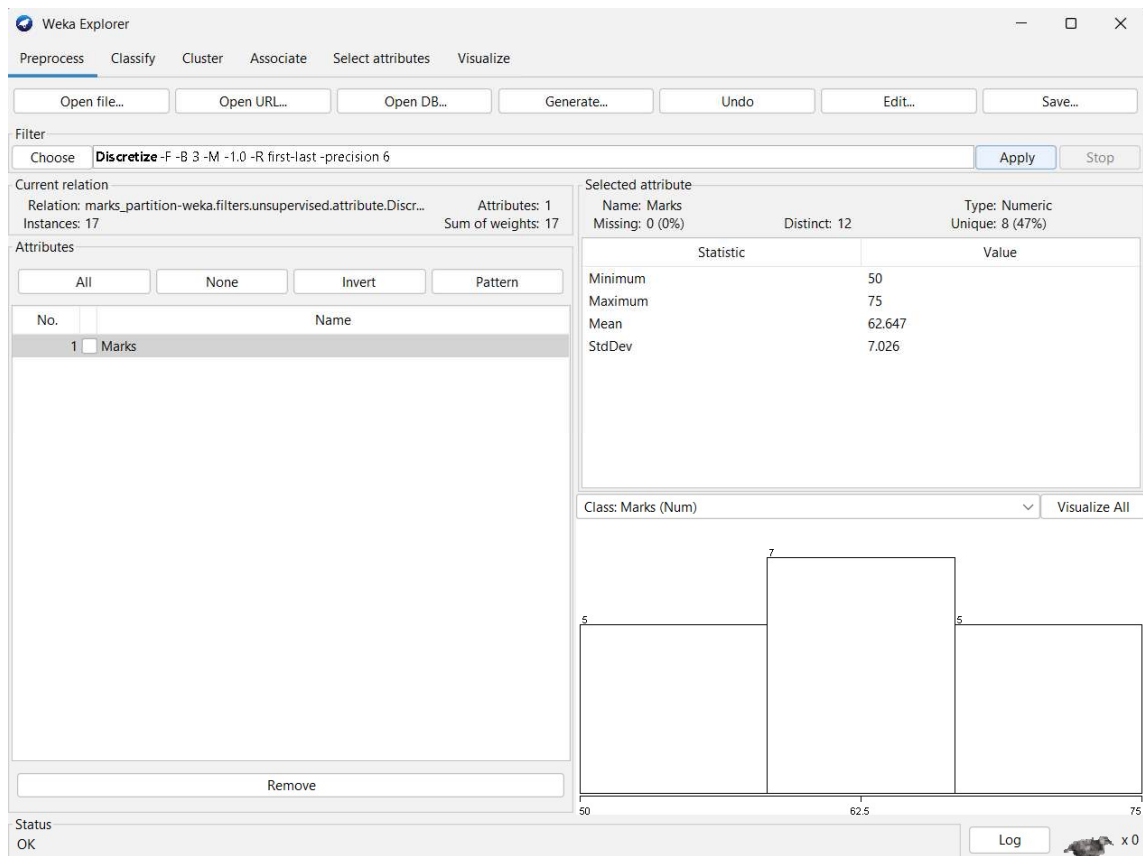
65

67

71

72

75



EXPERIMENT NO.:9

@relation gender\_decision

@attribute Height numeric

@attribute Weight numeric

@attribute Gender {Male,Female}

@data

185,70,Male

190,85,Male

175,90,Male

160,55,Female

170,60,Female

165,75,Female

182,82,Male

178,78,Female

185,95,Male

172,65,Female



**Weka Explorer**

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

**Test options**

☐ Use training set  
☐ Supplied test set Set...  
☒ Cross-validation Folds **10**  
☐ Percentage split % **66**  
 More options...

(Nom) Gender

Start Stop

Result list (right-click for options)

21:51:39 - trees.J48

**Classifier output**

Number of Leaves : 2  
Size of the tree : 3  
Time taken to build model: 0 seconds

=== Stratified cross-validation ===  
=== Summary ===

|                                  |          |    |   |
|----------------------------------|----------|----|---|
| Correctly Classified Instances   | 7        | 70 | % |
| Incorrectly Classified Instances | 3        | 30 | % |
| Kappa statistic                  | 0.4      |    |   |
| Mean absolute error              | 0.44     |    |   |
| Root mean squared error          | 0.5727   |    |   |
| Relative absolute error          | 80.6667  | %  |   |
| Root relative squared error      | 104.9974 | %  |   |
| Total Number of Instances        | 10       |    |   |

=== Detailed Accuracy By Class ===

|               | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC   | ROC Area | PRC Area | Class  |
|---------------|---------|---------|-----------|--------|-----------|-------|----------|----------|--------|
|               | 0.600   | 0.200   | 0.750     | 0.600  | 0.667     | 0.408 | 0.480    | 0.650    | Male   |
|               | 0.800   | 0.400   | 0.667     | 0.800  | 0.727     | 0.408 | 0.480    | 0.633    | Female |
| Weighted Avg. | 0.700   | 0.300   | 0.708     | 0.700  | 0.697     | 0.408 | 0.480    | 0.642    |        |

=== Confusion Matrix ===

```

a b  <-- classified as
3 2 | a = Male
1 4 | b = Female
  
```

Status: OK

Log x0

**Weka Explorer**

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **JRip -F 3 -N 2.0 -O 2 -S 1**

**Test options**

☐ Use training set  
☐ Supplied test set Set...  
☒ Cross-validation Folds **10**  
☐ Percentage split % **66**  
 More options...

(Nom) Gender

Start Stop

Result list (right-click for options)

21:51:39 - trees.J48  
21:52:43 - rules.JRip

**Classifier output**

(Height >= 182) => Gender=Male (4.0/0.0)  
=> Gender=Female (6.0/1.0)

Number of Rules : 2  
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===  
=== Summary ===

|                                  |        |    |   |
|----------------------------------|--------|----|---|
| Correctly Classified Instances   | 8      | 80 | % |
| Incorrectly Classified Instances | 2      | 20 | % |
| Kappa statistic                  | 0.6    |    |   |
| Mean absolute error              | 0.36   |    |   |
| Root mean squared error          | 0.4817 |    |   |
| Relative absolute error          | 66     | %  |   |
| Root relative squared error      | 88.305 | %  |   |
| Total Number of Instances        | 10     |    |   |

=== Detailed Accuracy By Class ===

|               | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC   | ROC Area | PRC Area | Class  |
|---------------|---------|---------|-----------|--------|-----------|-------|----------|----------|--------|
|               | 0.800   | 0.200   | 0.800     | 0.800  | 0.800     | 0.600 | 0.640    | 0.740    | Male   |
|               | 0.800   | 0.200   | 0.800     | 0.800  | 0.800     | 0.600 | 0.640    | 0.740    | Female |
| Weighted Avg. | 0.800   | 0.200   | 0.800     | 0.800  | 0.800     | 0.600 | 0.640    | 0.740    |        |

=== Confusion Matrix ===

```

a b  <-- classified as
4 1 | a = Male
1 4 | b = Female
  
```

Status: OK

Log x0

EXPERIMENT NO.:10

@relation electronics\_transactions

@attribute SONY {t,f}

@attribute BPL {t,f}

@attribute LG {t,f}

@attribute SAMSUNG {t,f}

@attribute ONIDA {t,f}

@data

t,t,t,f,f

f,t,f,t,f

f,t,f,f,t

t,t,f,t,f

t,f,f,f,t

f,t,f,f,t

t,f,f,f,t

t,t,t,f,t

t,t,f,f,t

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1**

Start Stop

Result list (right-click for ...)

21:56:08 - Apriori

Associator output

```

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.35 (3 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 13

Generated sets of large itemsets:

Size of set of large itemsets L(1): 7

Size of set of large itemsets L(2): 13

Size of set of large itemsets L(3): 9

Size of set of large itemsets L(4): 2

Best rules found:

1. ONIDA=t 6 ==> SAMSUNG=f 6 <conf:(1)> lift:(1.29) lev:(0.15) [1] conv:(1.33)
2. LG=f ONIDA=t 5 ==> SAMSUNG=f 5 <conf:(1)> lift:(1.29) lev:(0.12) [1] conv:(1.11)
3. LG=f SAMSUNG=f 5 ==> ONIDA=t 5 <conf:(1)> lift:(1.5) lev:(0.19) [1] conv:(1.67)
4. SONY=t ONIDA=t 4 ==> SAMSUNG=f 4 <conf:(1)> lift:(1.29) lev:(0.1) [0] conv:(0.89)
5. BPL=t ONIDA=t 4 ==> SAMSUNG=f 4 <conf:(1)> lift:(1.29) lev:(0.1) [0] conv:(0.89)
6. SONY=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
7. SONY=f 3 ==> LG=f 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
8. ONIDA=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
9. SONY=f LG=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
10. SONY=f BPL=t 3 ==> LG=f 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)

```

Status OK

Log x0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1**

Start Stop

Result list (right-click for ...)

21:56:08 - Apriori

21:56:34 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:    electronics_transactions
Instances:   9
Attributes:  5
             SONY
             BPL
             LG
             SAMSUNG
             ONIDA

=== Associator model (full training set) ===

FPGrowth found 8 rules (displaying top 8)

1. [BPL=f]: 2 ==> [SAMSUNG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
2. [SONY=f]: 3 ==> [LG=f]: 3 <conf:(1)> lift:(1.29) lev:(0.07) conv:(0.67)
3. [BPL=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
4. [SAMSUNG=f, SONY=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
5. [BPL=f]: 2 ==> [SAMSUNG=f, LG=f]: 2 <conf:(1)> lift:(1.8) lev:(0.1) conv:(0.89)
6. [SAMSUNG=f, BPL=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
7. [LG=f, BPL=f]: 2 ==> [SAMSUNG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
8. [SONY=f, ONIDA=f]: 1 ==> [LG=f]: 1 <conf:(1)> lift:(1.29) lev:(0.02) conv:(0.22)

```

Status OK

Log x0

EXPERIMENT NO.:12

@relation strike\_rate

@attribute StrikeRate numeric

@data

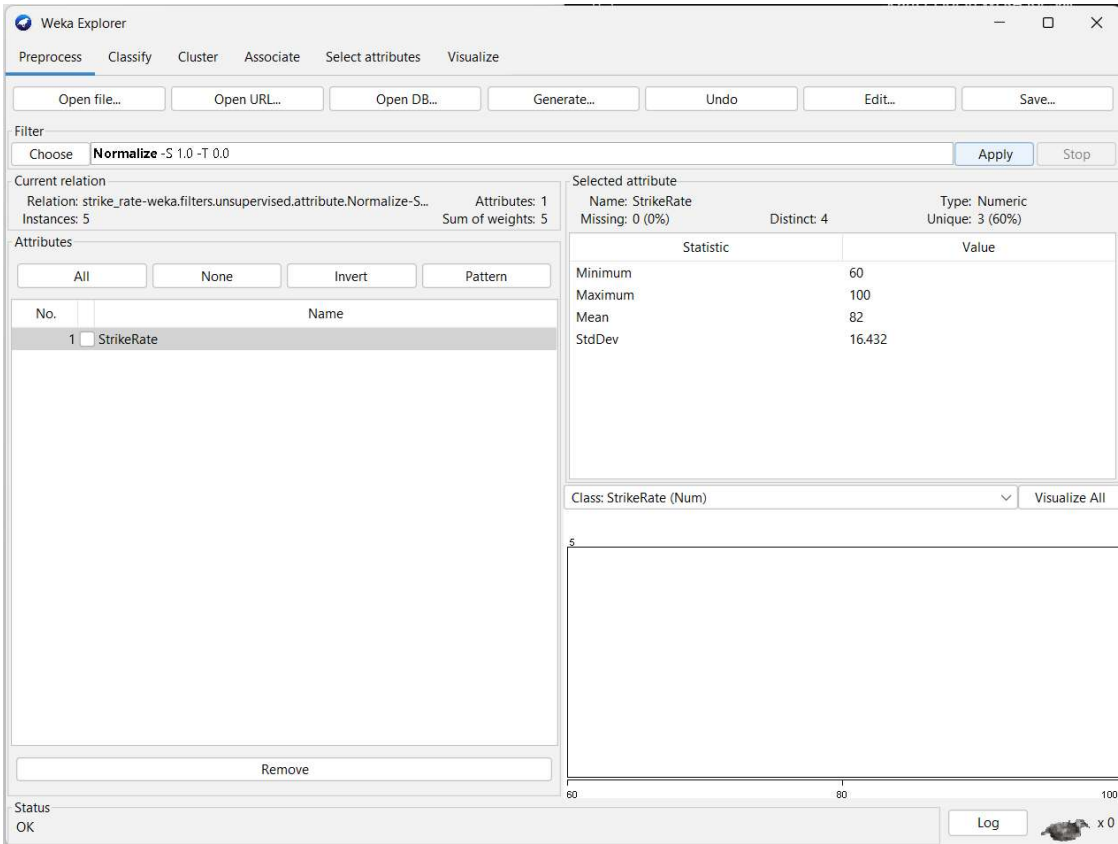
100

70

60

90

90



Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

FilterChooseStandardizeApplyStop

Current relation  
Relation: strike\_rate-weka.filters.unsupervised.attribute.Normalize-S...  
Instances: 5  
Attributes: 1  
Sum of weights: 5

Attributes  
AllNoneInvertPattern  
No. Name  
1 StrikeRate  
Remove

Selected attribute  
Name: StrikeRate  
Missing: 0 (0%)  
Distinct: 4  
Type: Numeric  
Unique: 3 (60%)

| Statistic | Value  |
|-----------|--------|
| Minimum   | 60     |
| Maximum   | 100    |
| Mean      | 82     |
| StdDev    | 16.432 |

Class: StrikeRate (Num)Visualize All

5  
6080100

Status  
OKLogx 0